



Time of death estimation using forehead temperature measurements

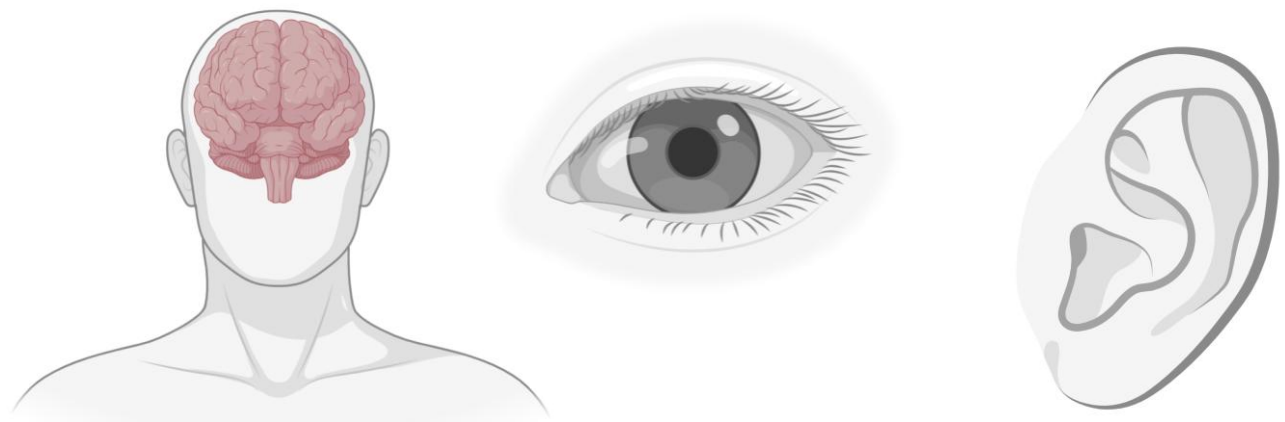
Bedarf Lennart, Gerlach Kathrin, Wittig Holger, Scheurer Eva, Berger Celine, Lenz Claudia

Institute of Forensic Medicine, Department of Biomedical Engineering, University of Basel, Switzerland
Institute of Forensic Medicine, Health Department Basel-Stadt, Basel, Switzerland

BACKGROUND

Time of death (ToD) estimation: critical for event reconstruction and legal investigations

- Current standard: **Henssge nomogram** based on body core temperature^{1, 2}
- Challenges: clothing, individual body composition, varying ambient temperature and invasive procedure^{1, 3-6}
- Alternative temperature sites^{7-9, 16}



Brain, eye, auditory canal



Forehead

→ **Forehead offers several advantages^{10, 11}:**

- Stable anatomy
- High thermal conductivity
- Low fat influence
- Correlation with brain temperature

Goal

Reliable and non-invasive tool for ToD estimation using forehead temperature

METHODS

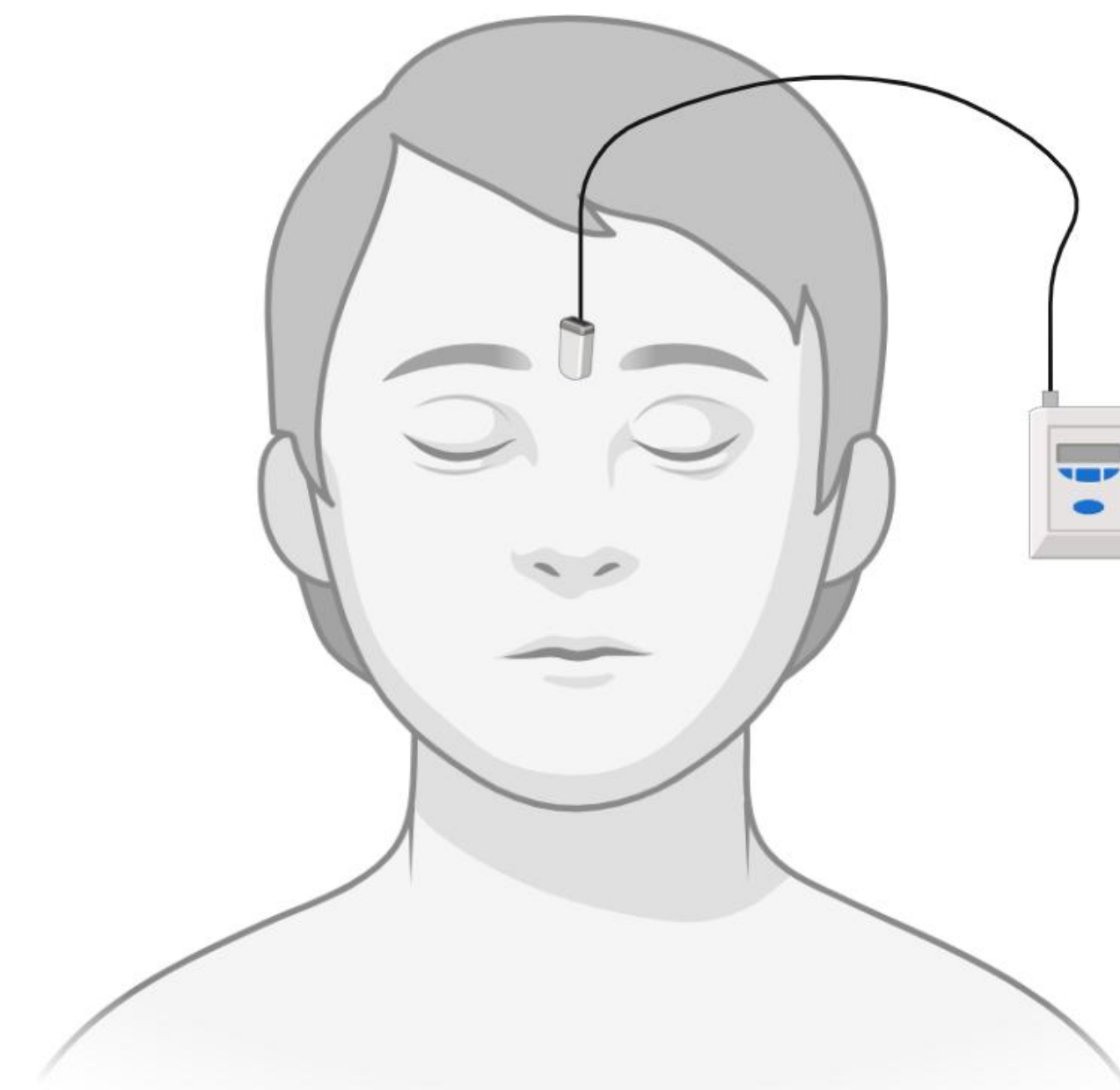


Fig. 1: Placement of the temperature probe¹⁶.

- 31 deceased individuals with known ToD
- Environment temperature at ToD: ~ 20 - 22 °C
- Continuous temperature measurement

→ Transport to cooling chamber at local institute of forensic medicine (2.5 °C)

Fitting: double-exponential equation based on **Marshall-Hoare-Henssge framework¹²⁻¹⁵:**

$$\frac{T(t) - T_a}{T_0 - T_a} = \alpha e^{-\beta t} - \gamma e^{-\delta t}$$

RESULTS

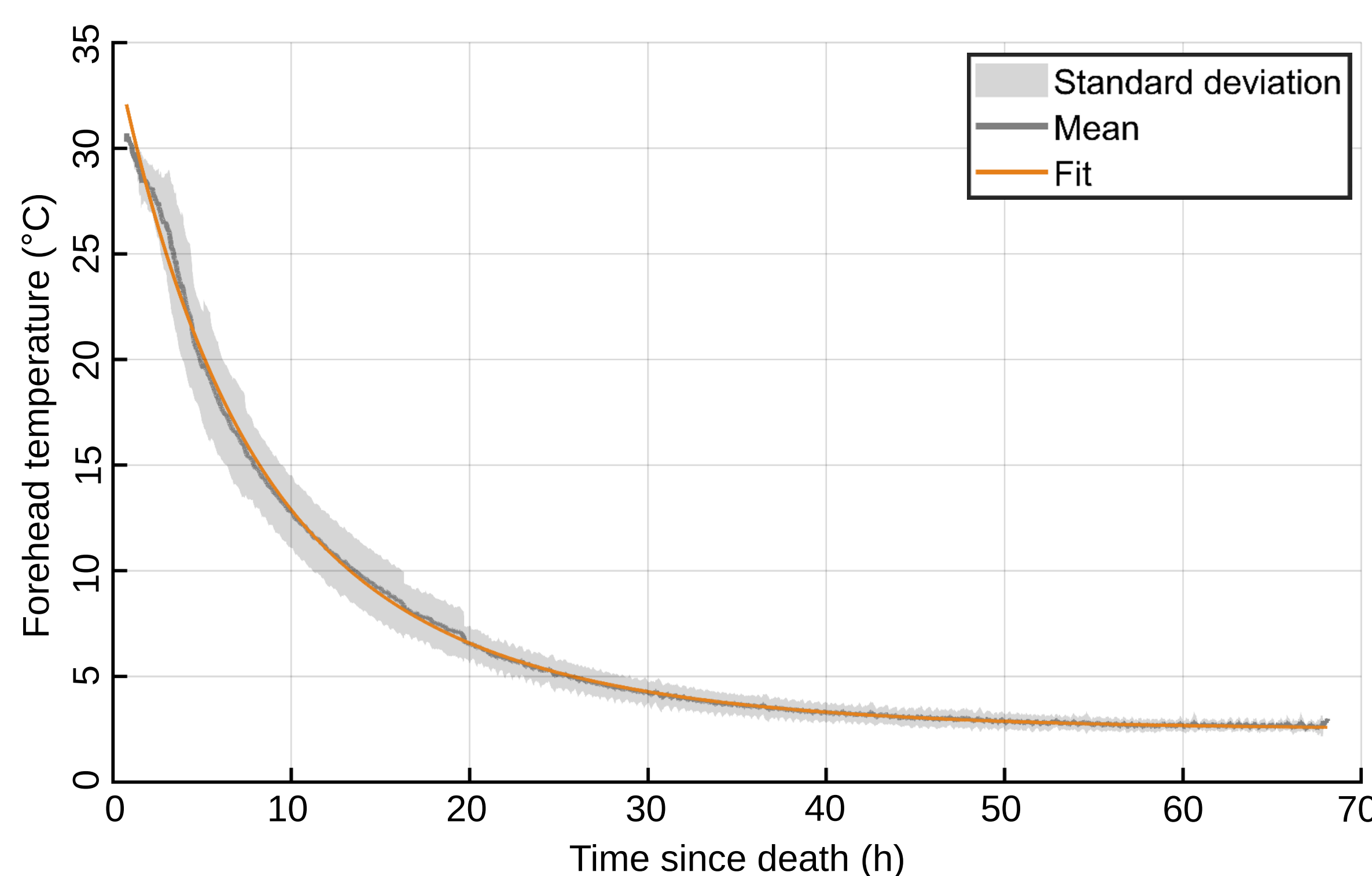


Fig. 2: Averaged forehead temperature (grey line), corresponding standard deviation (shaded grey area), and fit function (orange).

α	β	γ	δ
0.5101	0.1829	-0.5477	0.0763
± 0.00520	± 0.0013	± 0.0055	± 0.0003

Tab. 1: Resulting fit parameters and corresponding standard deviations.

RMSE: 0.25 °C

R²: 0.999

- No observable temperature plateau

- Variability:
 - Low inter-individual variability
 - Consistent cooling patterns across subjects
 - Minimal effect of body composition or fat distribution

Strong agreement of observed vs. modelled temperature curves

EXAMPLE CALCULATION

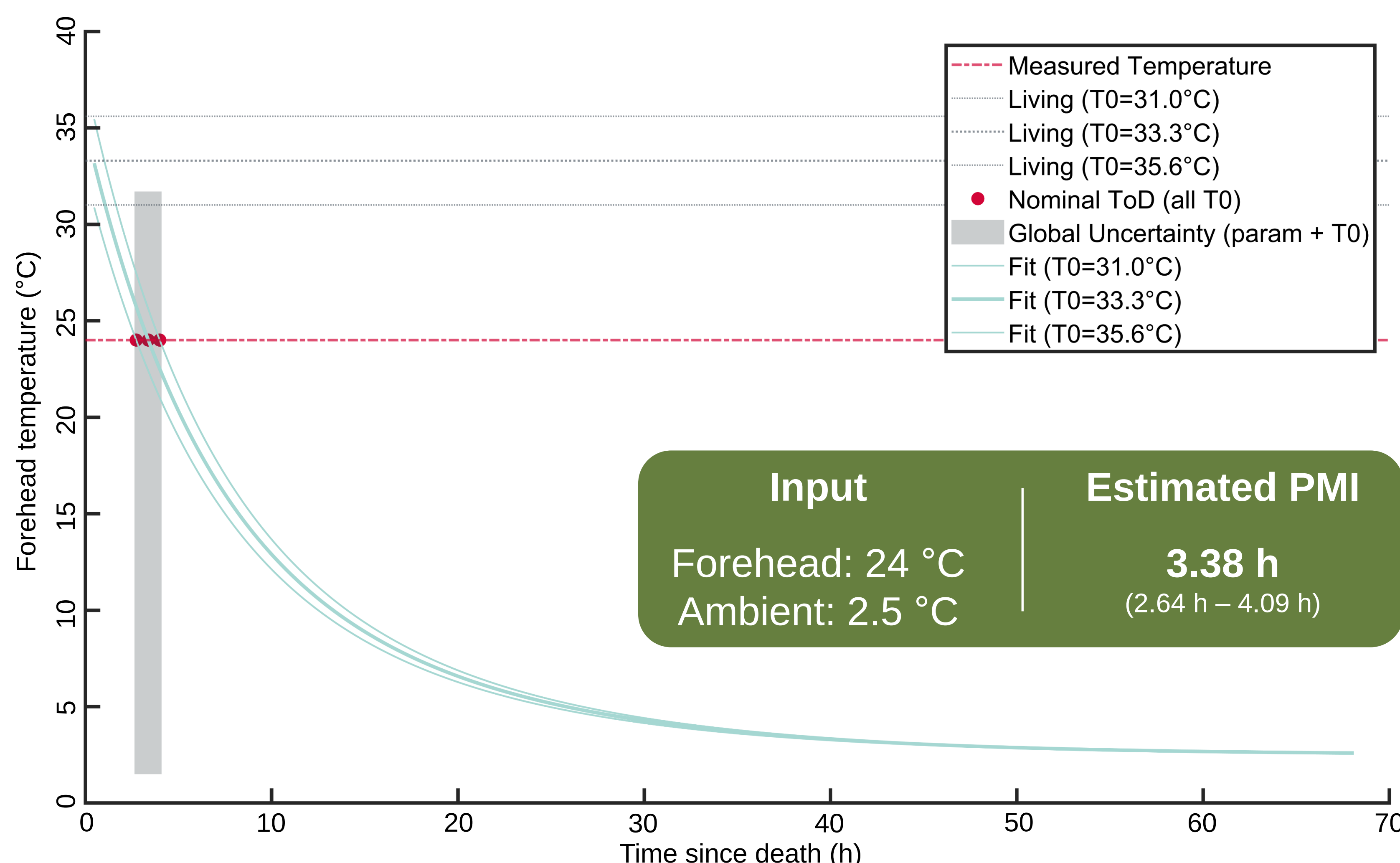


Fig. 3: Graphical depiction of the estimated ToD. Grey area: uncertainty range.

DISCUSSION

- Low inter-individual variability
- Low inter-operator variability
- Non invasive
- Limitations: controlled indoor conditions

- Future perspectives:
 - Validation under real-world forensic scenarios
 - Comparison with established methods
 - Integration into forensic workflows

Promising complementary tool to improve accuracy and practicality in ToD estimation



REFERENCES

1. Henssge, FSI 38 (1988) 209–236. [https://doi.org/10.1016/0379-0738\(88\)90168-5](https://doi.org/10.1016/0379-0738(88)90168-5).
2. Henssge et al., Z Rechtsmed 93 (1984) 1–22. <https://doi.org/10.1007/BF00202979>.
3. Al-Alousi, FSI 125 (2002) 237–244. [https://doi.org/10.1016/S0379-0738\(01\)00651-X](https://doi.org/10.1016/S0379-0738(01)00651-X).
4. Bartgis et al., J Therm Biol 62 (2016) 109–115. <https://doi.org/10.1016/j.jtherbio.2016.07.004>.
5. Madea et al., Handbook of Forensic Medicine (2014) 75–133. <https://doi.org/10.1002/9781118570654.ch7>.
6. Zhang et al., Heat Transfer Eng. 28 (2007) 335–342. <https://doi.org/10.1080/01457630601122823.12>.
7. Rutty, Forens Sci Med Pathol 1 (2005) 41–51. <https://doi.org/10.1385/FSMP.1.1.041>.
8. Kaliszán et al., Legal Medicine 38 (2019) 10–13. <https://doi.org/10.1016/j.legalmed.2019.03.004>.
9. Madea, CRC Press, 2023.
10. Berger et al., J Therm Biol 115 (2023) 103615. <https://doi.org/10.1016/j.jtherbio.2023.103615>.
11. Kameda, J. Perioper. Pract. 32 (2022) 142–148. <https://doi.org/10.1177/17504589211022314>.
12. Brown et al., Forensic Science 4 (1974) 125–133. [https://doi.org/10.1016/0300-9432\(74\)90093-4](https://doi.org/10.1016/0300-9432(74)90093-4).
13. Henßge, Z Rechtsmed 83 (1979) 49–67. <https://doi.org/10.1007/BF00201311>.
14. Eckl, Ludwig-Maximilians-Universität zu München, 2004.
15. Henssge, Z Rechtsmed 87 (1981) 147–178. <https://doi.org/10.1007/BF00204763>.
16. Figure created with BioRender.com
17. Bedarf et al., IJLM, 2026. <https://doi.org/10.1007/s00414-026-03736-x>
18. https://github.com/lenn19/ToD_temperature

